

The Sound of Serotonin

by Sonific-Eight

BACKGROUND

Mood regulation disorders
affect 1 in 5 adults in the UK



RESEARCH PROBLEM

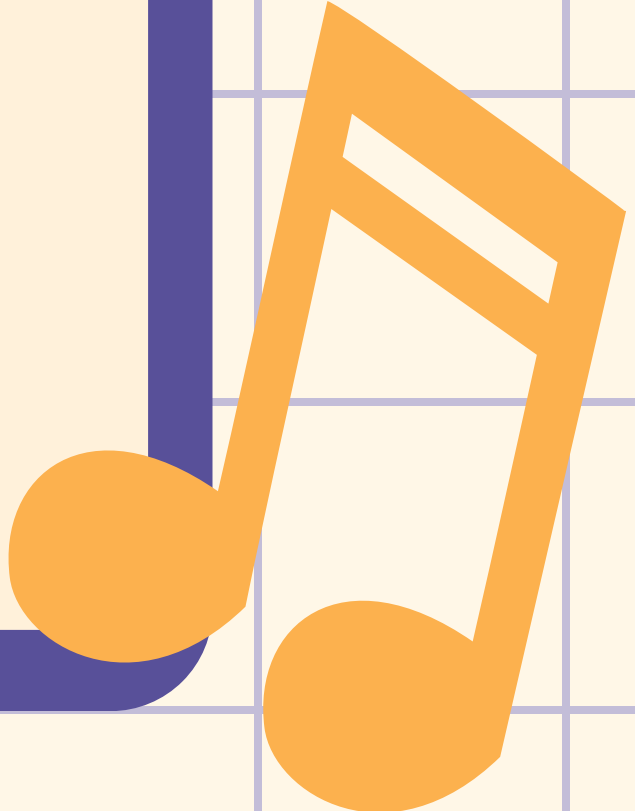
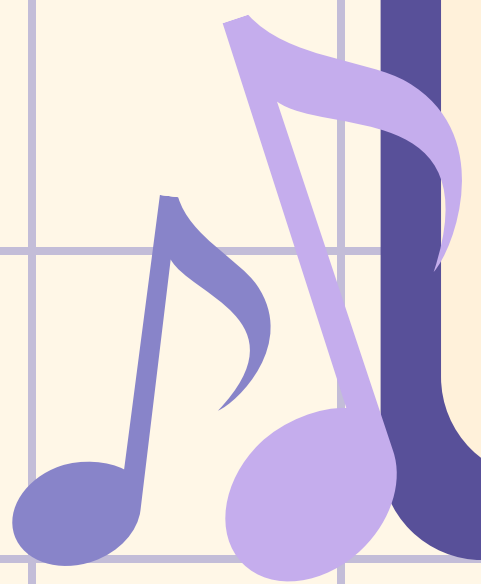
Meaningful access and interpretation to
serotonin data remains limited





OUR SOLUTION

**Our project transforms
spontaneous serotonin datasets
into structured musical output**



AIMS AND OUTCOMES

Project
Aims



Sonify
Serotonin
Signals



Improve
Accessibility

Outcomes



THE GREAT
EXHIBITION
ROAD FESTIVAL



Research
Publication



TASK ALLOCATION



Software Development



Iacopo



Dan



Javi



Ines

Design and Implementation



Lavinia



Yasmina

Logistical Planning

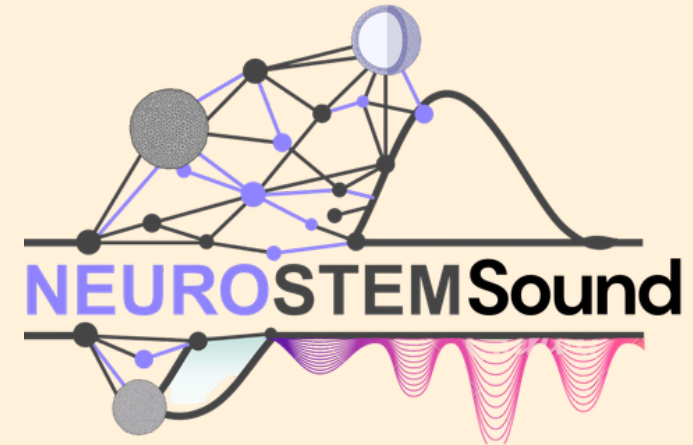
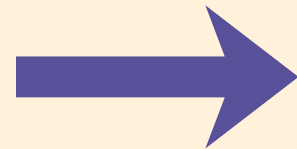
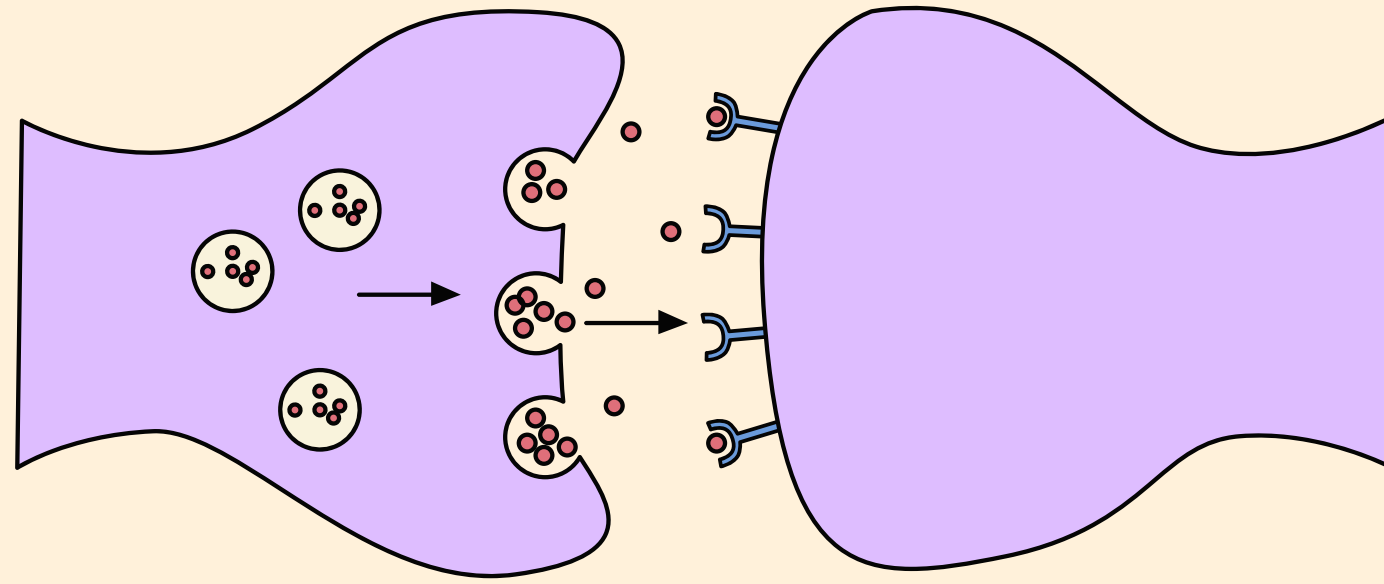


Hetal



Joe

NEUROSTEMSOUND

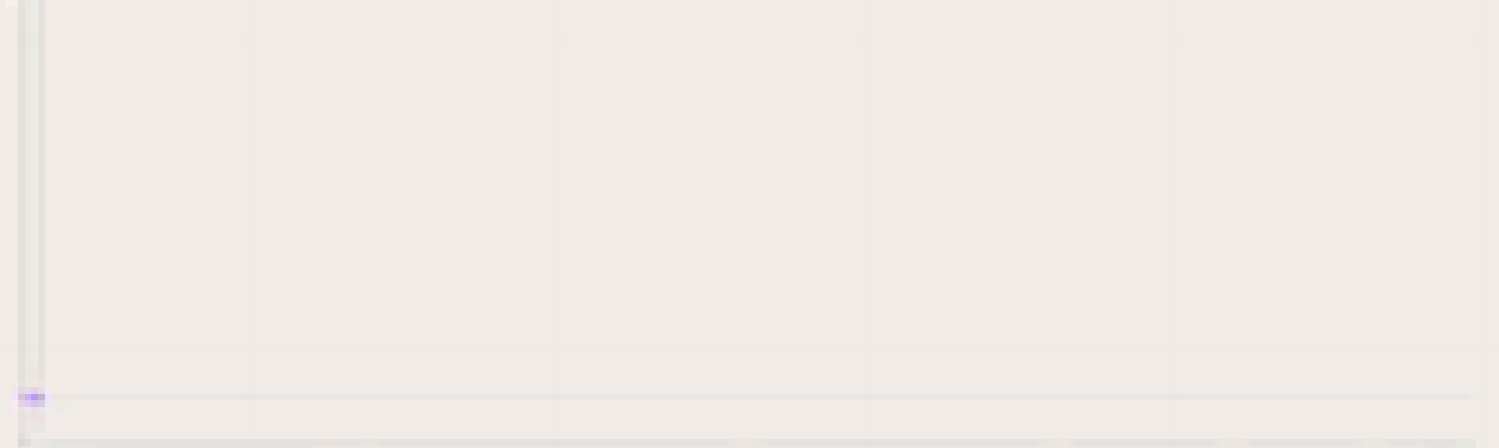


Action potential (voltage vs time)
mV



3

Concentration vs time
nM



3

SOFTWARE DEMO

NeuroStemVolt

Reverse Changes

231204_organoid3_FL_forskolin

58 post-drug_COLOR.txt

Previous

Next

Filter Options

Evaluate

Save Current Plots

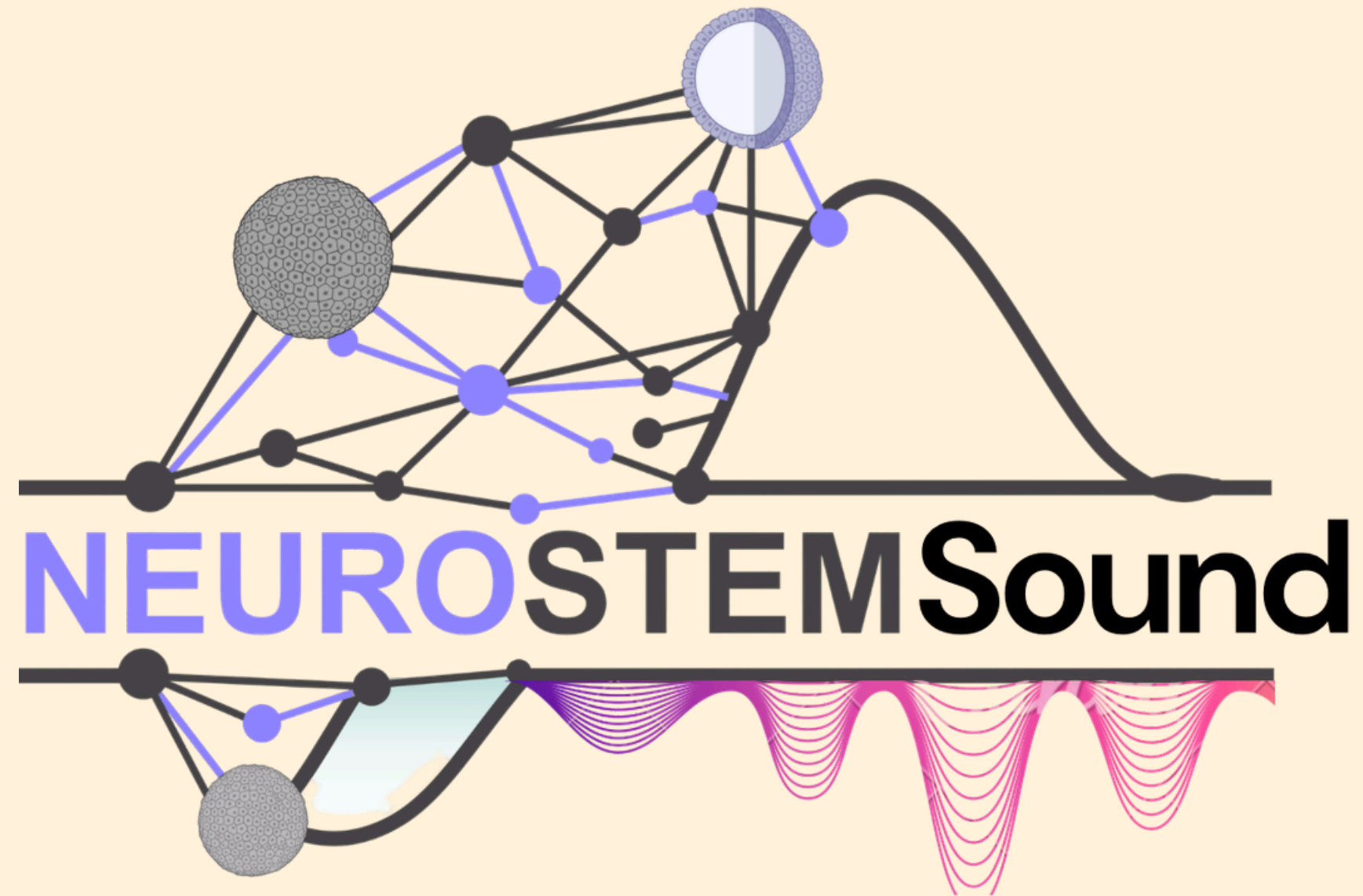
Export Current IT

Characteristic Extraction

Sonify

Export All ITs

Apply Peak Adjustment



SOFTWARE DEMO

1

Load neurotransmitter file

231204_organoid3_FL_forskolin

58 post-drug_COLOR.txt

Previous

Next

Filter Options

Evaluate

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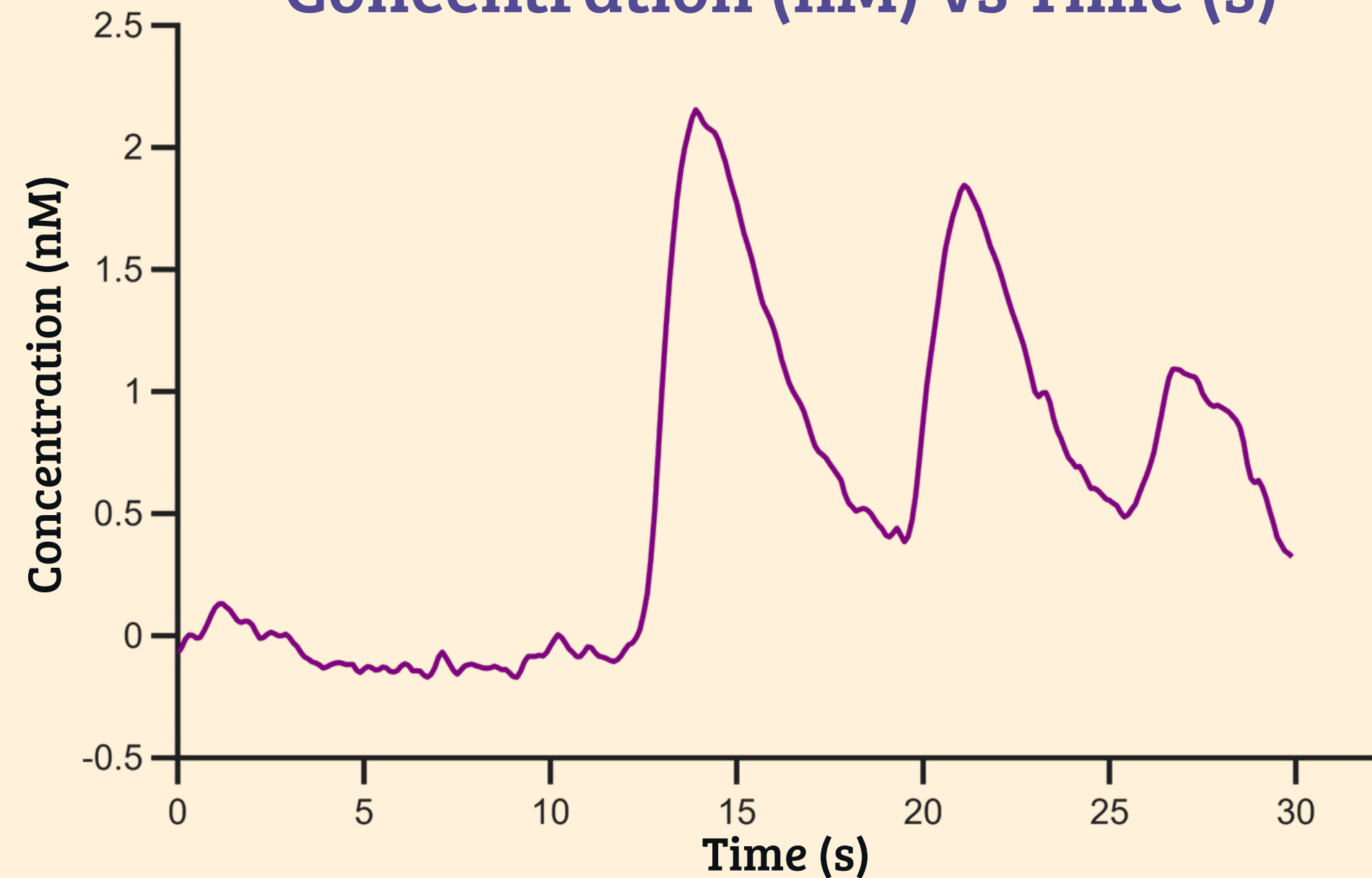
Characteristic Extraction

Sonify

Export All ITs

Apply Peak Adjustment

Concentration (nM) vs Time (s)



SOFTWARE DEMO

1

Load neurotransmitter file

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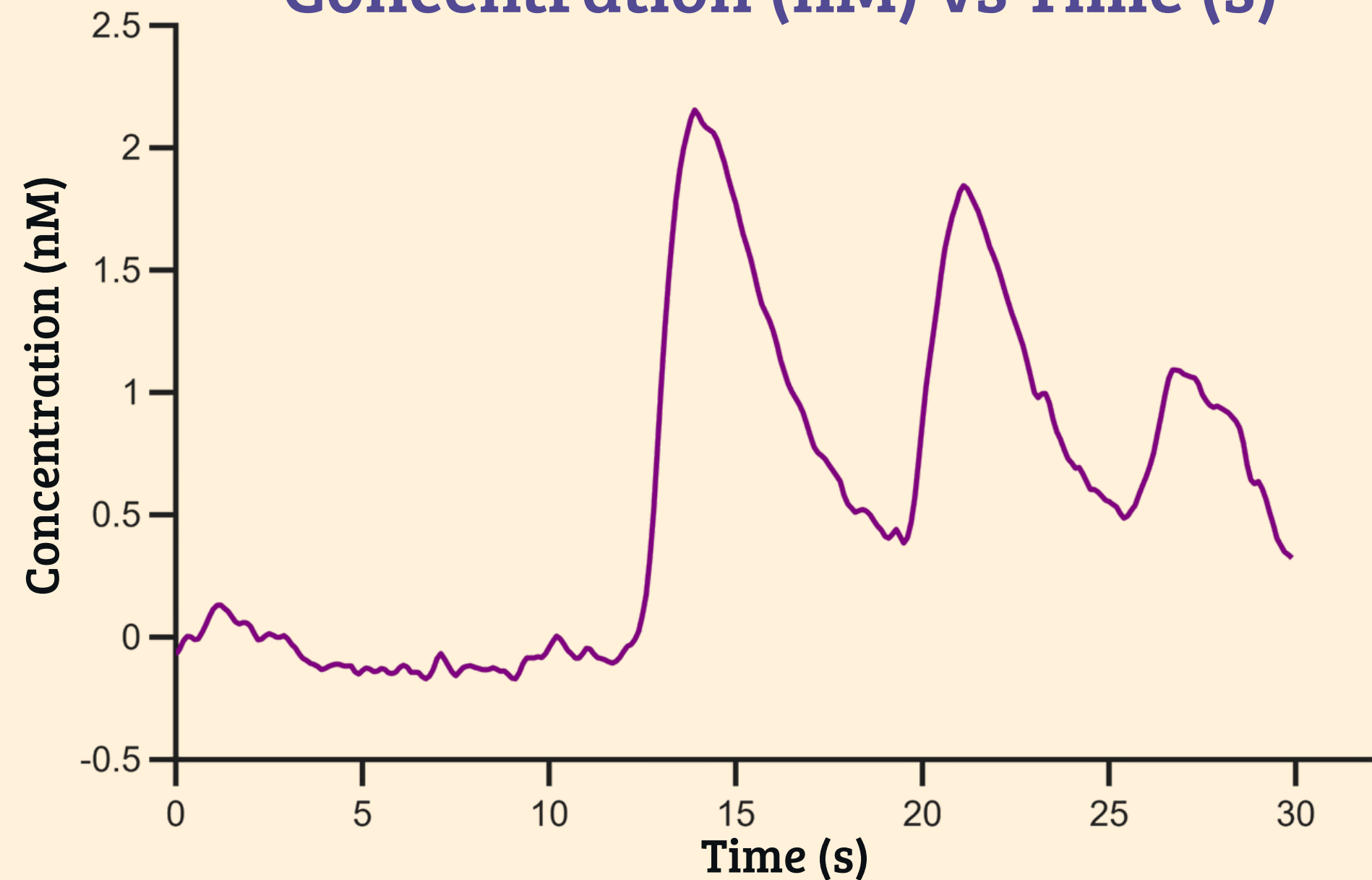
Export All ITs

Apply Peak Adjustment

2

Filter

Concentration (nM) vs Time (s)



SOFTWARE DEMO

1

Load neurotransmitter file

231204_organoid3_FL_forskolin
58 post-drug_COLOR.txt

Previous

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Characteristic Extraction

Sonify

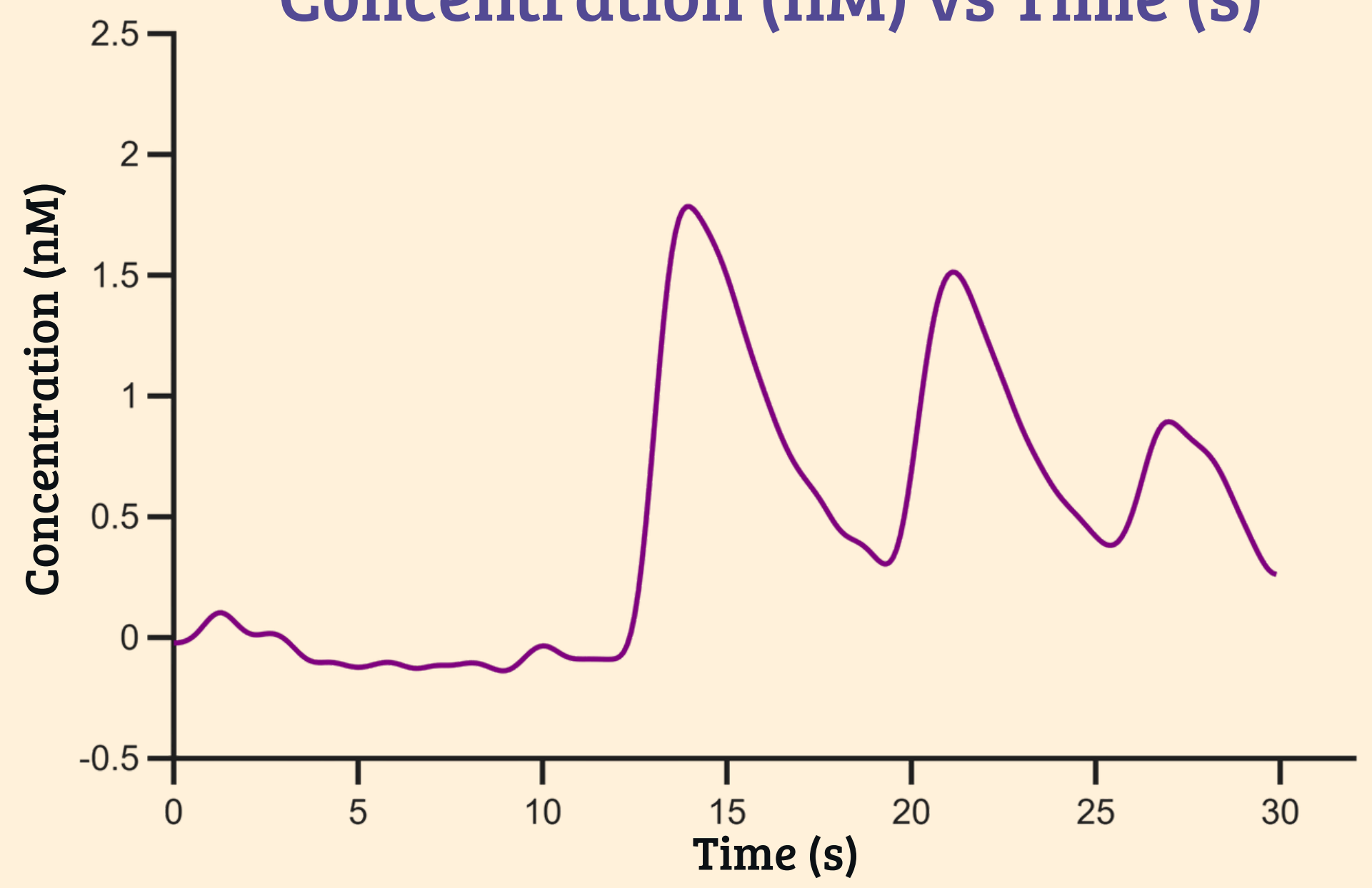
Export All ITs

Apply Peak Adjustment

2

Filter

Concentration (nM) vs Time (s)



SOFTWARE DEMO

1

Load neurotransmitter file

231204_organoid3_FL_forskolin

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Extract peak features

Evaluate

Save Current Plots

Export Current IT

Characteristic Extraction

Sonify

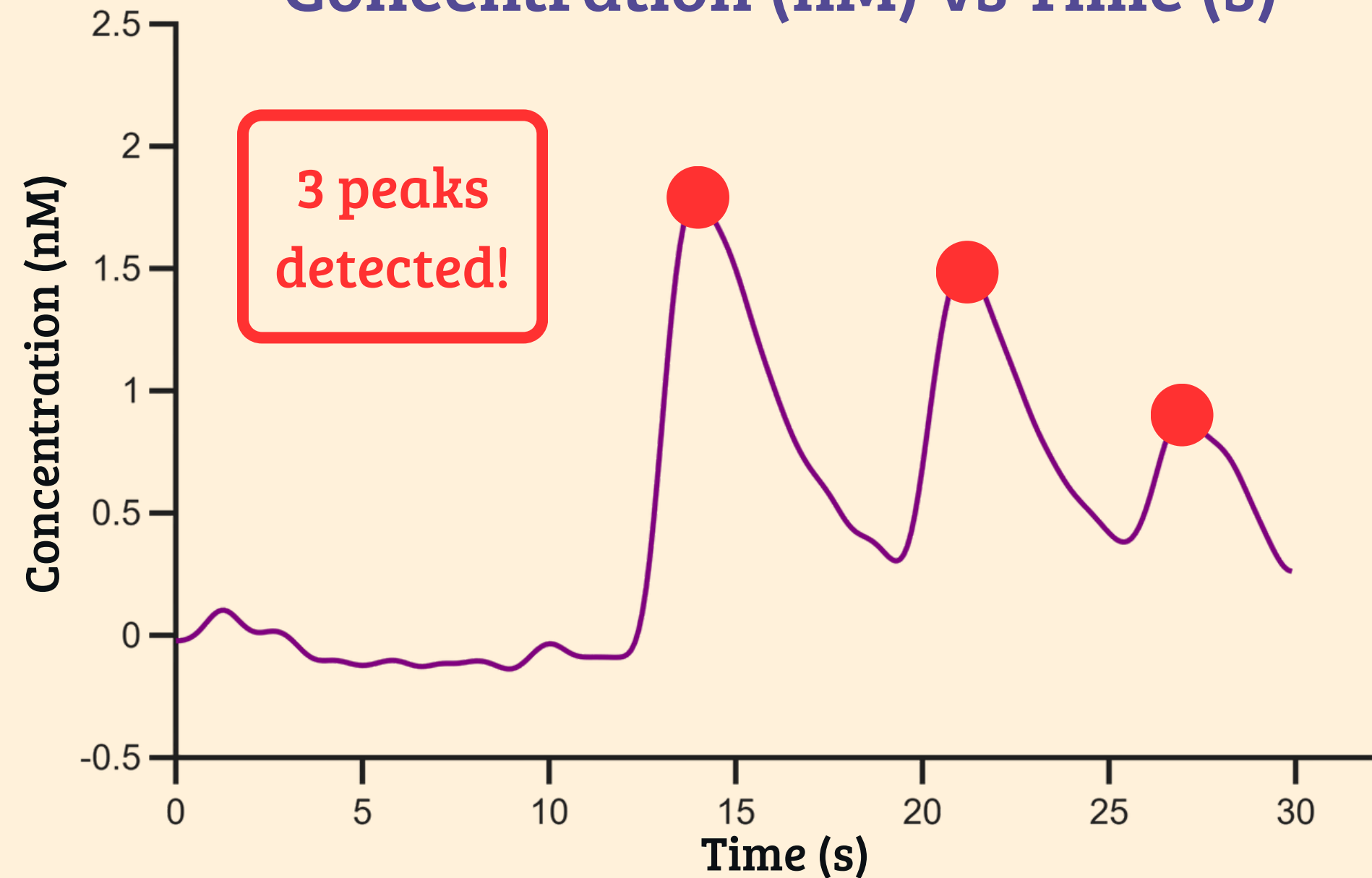
Export All ITs

Apply Peak Adjustment

2

Filter

Concentration (nM) vs Time (s)



SOFTWARE DEMO

1

Load neurotransmitter file

231204_organoid3_FL_forskolin
58 post-drug_COLOR.txt

Previous

Next

3

Extract peak features

Filter Options

Evaluate

Save Current Plots

Export Current IT

Characteristic Extraction

Sonify

Export All ITs

4

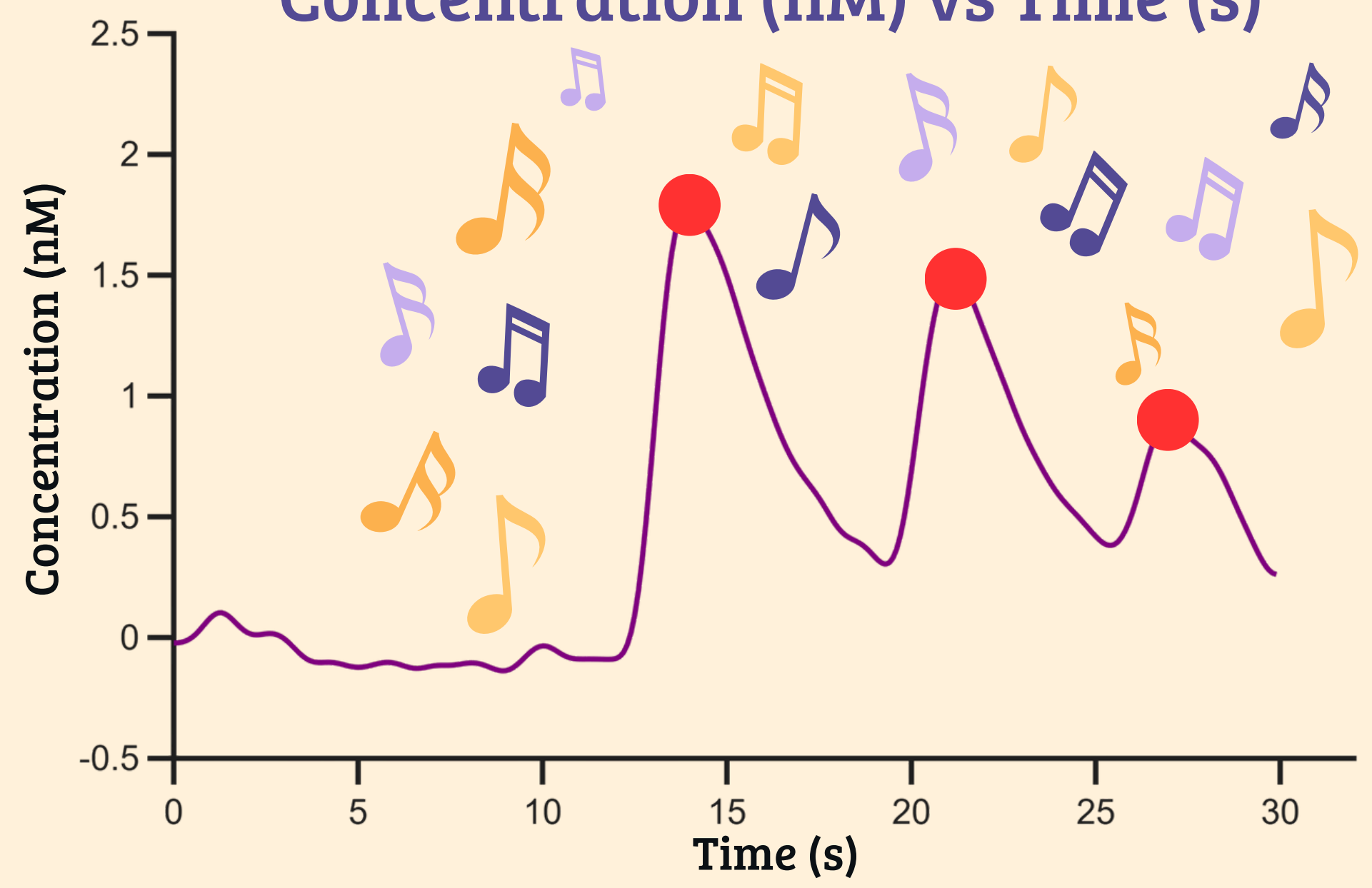
Sonify

Apply Peak Adjustment

2

Filter

Concentration (nM) vs Time (s)

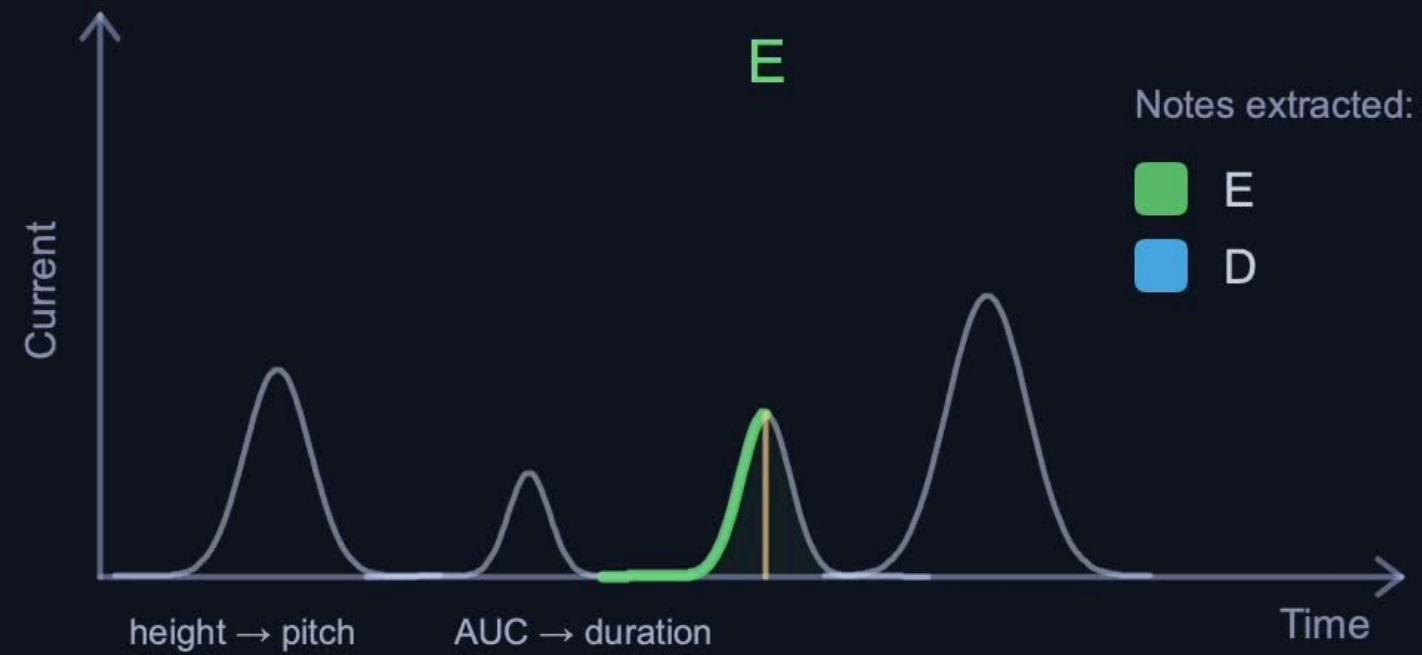


SONIFICATION

SONIFICATION

SONIFICATION

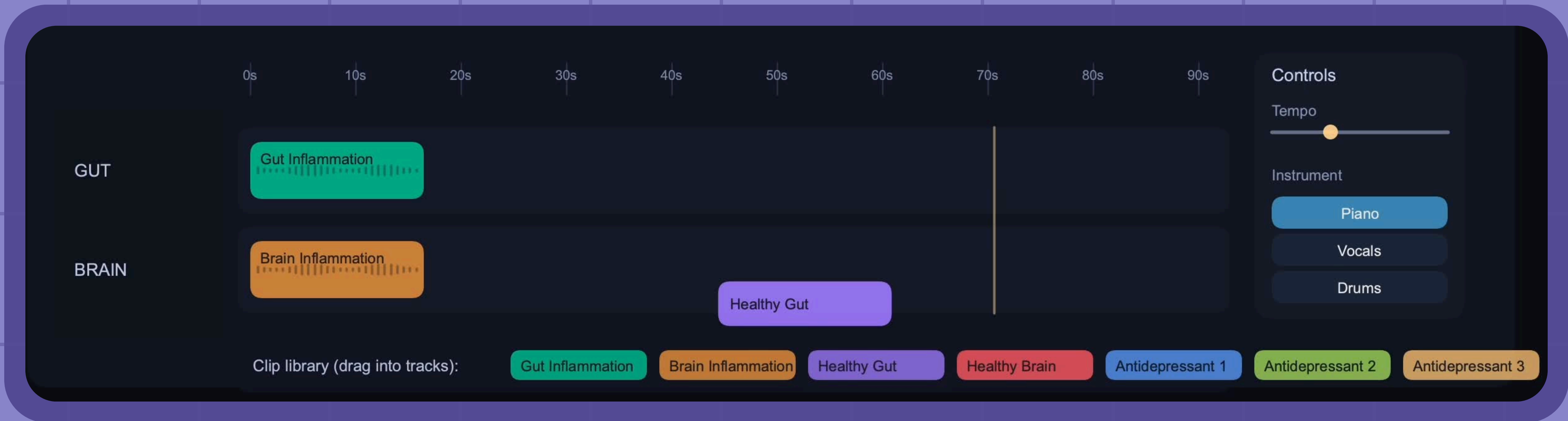
Sequential feature extraction (peak → pitch/duration)



Notes



SONIFICATION



WHAT ARE OUR BIGGEST RISKS?

	Potential risk	S (1-10)	L (1-10)	$S \times L$ (1-100)
1	Demo day failure	10	5	50
2	Reproducibility issues	9	5	45

HOW DO WE TACKLE THESE?

Mitigation

S
(1-10)

L
(1-10)

$S \times L$
(1-100)

1

- Backup export
- No last minute changes

5

3

15

2

- Use one fixed environment
- Choose demo laptop early

8

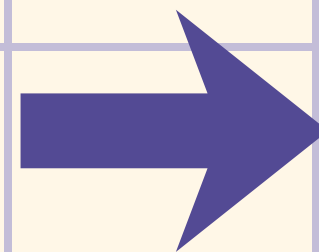
3

24

CONTINGENCY PLANNING

Average risk

29



Mitigated average risk

17

PROJECT SCHEDULE

Oct. – Dec.

Jan. – Mar.

Apr. – Jun.

Research

Prototyping software

Software refinement

Testing & report writing

Project Pitch

Plan B

Plan C

THANK YOU!